Project Report

DATA ANALYTICS DASHBOARD USING STREAMLIT

Ferry Capacity Utilization & Operational Efficiency Analytics System

Submitted By

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TechnologyUsed

- Python
- Pandas
- Plotly
- NumPy
- Seaborn
- Streamlit

Project Domain

Transportation Analytics

Submitted to

Unified mentor private limited

Abstract

This project focuses on analyzing ferry ticket sales and redemption data to understand how efficiently ferry services are being used. The main goal is to identify busy periods, low-activity periods, and overall usage patterns of ferry operations. By studying historical data collected at 15-minute intervals, the project helps in measuring capacity utilization and detecting operational inefficiencies.

The analysis includes examining passenger activity based on time, day, and season. Key performance indicators such as capacity utilization, congestion levels, and idle-capacity periods are used to evaluate ferry operations. An interactive Streamlit dashboard is also developed to display important insights through charts, graphs, and heatmaps.

The results of this project can help ferry operators make better decisions regarding scheduling, resource allocation, and service planning. Overall, the system provides a simple and effective way to improve operational efficiency and support data-driven decision-making in ferry transportation services.

Introduction

Ferry transportation is an important part of public transport in Toronto. Ferries operate from the Jack Layton Ferry Terminal and provide transportation to Centre Island, Hanlan's Point, and Ward's Island. Every day, thousands of passengers use ferry services for travel, tourism, and recreation.

Managing ferry operations efficiently is important because ferries have fixed seating capacities and require staff and operational resources. During some periods, ferries may become overcrowded, while during other periods they may operate with very few passengers. These situations can lead to operational inefficiencies and increased costs.

This project aims to analyse ferry ticket sales and redemption data to understand how ferry services are being utilized and to identify opportunities for improving operational efficiency.

Problem Statement

Despite the availability of detailed ticket sales and redemption data, ferry operations lack a structured analytical framework to:

- Measure capacity utilization patterns.
- Identifying over-utilization and under-utilization windows.
- Quantify operational inefficiencies during off-peak periods.
- Support evidence-based resource allocation decisions.

Objectives

Primary Objectives

- Measure ferry capacity utilization using ticket activity data.
- Identify inefficiencies in ferry deployment across time intervals.
- Detect congestion-prone and idle-capacity periods.

Secondary Objectives

- Support operational cost optimization.
- Improve passenger comfort and safety.
- Enable strategic planning for seasonal operations.

Dataset Description

Column Name	Description
_id	Unique record identifier
Timestamp	Date and time of record
Sales Count	Number of tickets sold
Redemption Count	Number of tickets used

Analytical Methodology

Data Preparation

The dataset is loaded and cleaned before analysis. The timestamp column is converted into a date and time format so that trends can be studied over different periods.

Additional features such as year, month, day, and season are extracted from the timestamp.

Data Cleaning

The following checks are performed:

- Missing values
- Duplicate records
- Negative values
- Unusual spikes in activity

Cleaning the data ensures accurate analysis.

Feature Engineering

New metrics are created to better understand ferry operations:

Total Activity

Total Activity = Sales Count + Redemption Count

This represents the overall passenger activity during a time interval.

Capacity Utilization

Capacity utilization measures how busy ferry operations are compared to the highest observed activity level.

Congestion Indicator

Intervals with very high activity are marked as congestion periods.

Idle Capacity Indicator

Intervals with very low activity are marked as idle-capacity periods.

Data Analysis

Daily and Hourly Trends

Passenger activity is analysed across different times of the day to identify peak and off-peak periods.

Weekday vs Weekend Analysis

Activity levels are compared between weekdays and weekends to understand travel behavior.

Seasonal Analysis

Passenger demand is compared across seasons:

- Spring
- Summer
- Winter

This helps identify periods of high and low ferry utilization.

Key Performance Indicators (KPIs)

The following KPIs are used to evaluate operational efficiency:

Capacity Utilization Ratio

Measures how effectively ferry services are being used.

Congestion Pressure Index

Measures the percentage of intervals with very high passenger activity.

Idle Capacity Percentage

Measures the percentage of intervals with very low passenger activity.

Peak Strain Duration

Measures how long congestion continues during busy periods.

Operational Variability Score

Measures how much passenger activity changes over time.

Dashboard Development

A Streamlit dashboard is developed to provide interactive visualizations and operational insights.

Dashboard Features

- KPI summary cards
- Capacity utilization timeline
- Congestion heatmaps
- Idle-capacity heatmaps
- Seasonal comparison charts
- Interactive filters

Users can select specific dates, seasons, and time intervals for analysis.

Expected Outcomes

The project is expected to provide answers to questions such as:

- What are the busiest ferry operating periods?
- Which time intervals experience congestion?
- Which periods have low utilization?
- How does utilization change across seasons?
- What operational improvements can be made?

Recommendations

Based on the analysis, the following recommendations can be made:

- Increase ferry frequency during high-demand periods.
- Reduce unnecessary operations during low-demand periods.
- Adjust staffing levels according to passenger activity.
- Use seasonal trends for future planning.
- Continuously monitor operational performance through dashboards.

REFERENCES

- Toronto Government Parks, Forestry & Recreation. Ferry Capacity Utilization & Operational Efficiency Analytics System Project Description.
- McKinney, W. (2022). Python for Data Analysis. O'Reilly Media.
- VanderPlas, J. (2016). Python Data Science Handbook. O'Reilly Media.
- Pandas Documentation.
- NumPy Documentation.
- Streamlit Documentation.
- Plotly Documentation.

Appendix

Software and Tools Used

- Python
- Pandas
- NumPy
- Plotly
- Streamlit
- Jupyter Notebook
- Visual Studio Code (VS Code)

SOURCE CODE SNAPSHOT

```
capacity = df["Redemption Count"].quantile(0.95)

df["utilization"] = (
    df["Redemption Count"] / capacity
).clip(upper=1)
```

```
df['utilization']= 1-df['Idle_Capacity']
```

```
df["Congestion"] = df["Redemption_Pressure_Ratio"] > 0.85
```

```
(df["utilization"]< 0.2).mean() * 100
```

```
peak = df["Total_Activity_Load"] > 0.9
groups = (peak != peak.shift()).cumsum()
peak_lengths = peak.groupby(groups).sum()
peak_duration_intervals = peak_lengths.max()
peak_duration_minutes = peak_duration_intervals * 15
print("Peak Strain Duration (minutes):", peak_duration_minutes)
```

Conclusion

This project focuses on analysing ferry ticket sales and redemption data to evaluate operational efficiency. By studying passenger activity patterns, congestion periods, and idle-capacity windows, the system helps identify opportunities for improving ferry operations.

The Streamline dashboard provides a user-friendly platform for monitoring ferry utilization and supporting data-driven decision-making. Overall, the project contributes to better resource management, improved passenger experience, and more efficient ferry transportation services.